

CUSTOMER CHURN ANALYSIS

Data Cleaning, Feature Engineering, Analysis & Visualization Report

1. Executive Summary

This report documents the customer churn analysis performed in the jupyter notebook. The workflow combines customer, subscription and support data, cleans and standardizes fields, creates a churn flag and churn-risk category, and then evaluates churn across plans, states and months. The notebook also explores relationships among churn score, churn flag, escalations and churn risk.

2. Data Sources and Data Preparation

The report has three tables: db_customer, db_subscription and db_support. The tables contain customer attributes, subscription information and support/complaint information respectively.

Table	Key fields
db_customer	Customer ID, name, country, state, gender, DOB, interests, pincode
db_subscription	Subscription dates, plan/contract type, cancellation, charges, CLTV, churn score
db_support	Customer ID, complaint date, escalations, CSAT score, comments

2.1 Cleaning steps

- Renamed the customer table's name field to customer_name.
- Dropped interests and pincode from the customer data.
- Converted dob to datetime.
- Standardized gender values by replacing Men with Male and Women with Female.
- Filled missing country values by mapping state to a known country value.
- Converted subscription_start_date, renewal_date and cancellation_date to datetime.
- Removed unused support fields col_1 and comment and converted complaint_date to datetime.
- Created churn_flag from cancellation_date: customers with a cancellation date are assigned 1; otherwise 0.

2.2 Support-data duplicate handling

The support table contains repeated customer IDs. Initially complaint_count per customer was calculated, then sorts by complaint_date and keeps the latest support record per customer before the final merge. This reduces the support table and prevents support duplicates from inflating the final customer-level dataset.

3. Feature Engineering

Feature	How it is created / used
churn_flag	1 when cancellation_date is present; otherwise 0
complaint_count	Number of support/complaint records associated with each customer
customer_age	Year-based age calculation from DOB
tenure_days	Service duration in days using cancellation date or current date
churn_risk	low if churn_score < 50; med if 50–69; high if churn_score >= 70

4. Key Business Metrics

Metric	Result	Interpretation from notebook
Churn rate	28.57%	Share of customers with churn_flag = 1
Retention rate	71.43%	100% minus churn rate
Avg Revenue per user	18.85	Mean monthly_charges across customers
Average tenure	1,532 days	Average calculated service duration
Revenue at risk	73.94 (Rs 'K')	Sum of monthly_charges for churned customers
Escalation rate	19.05%	Share of records with escalation = Y

Average complaints/user	0.43	Total complaints divided by unique customers
-------------------------	------	--

5. Churn Analysis by Plan Type

Calculation of churn rate as the mean of churn_flag within each plan type.

Plan type	Churn rate
Basic	60.00%
Premium	14.29%
Standard	22.22%

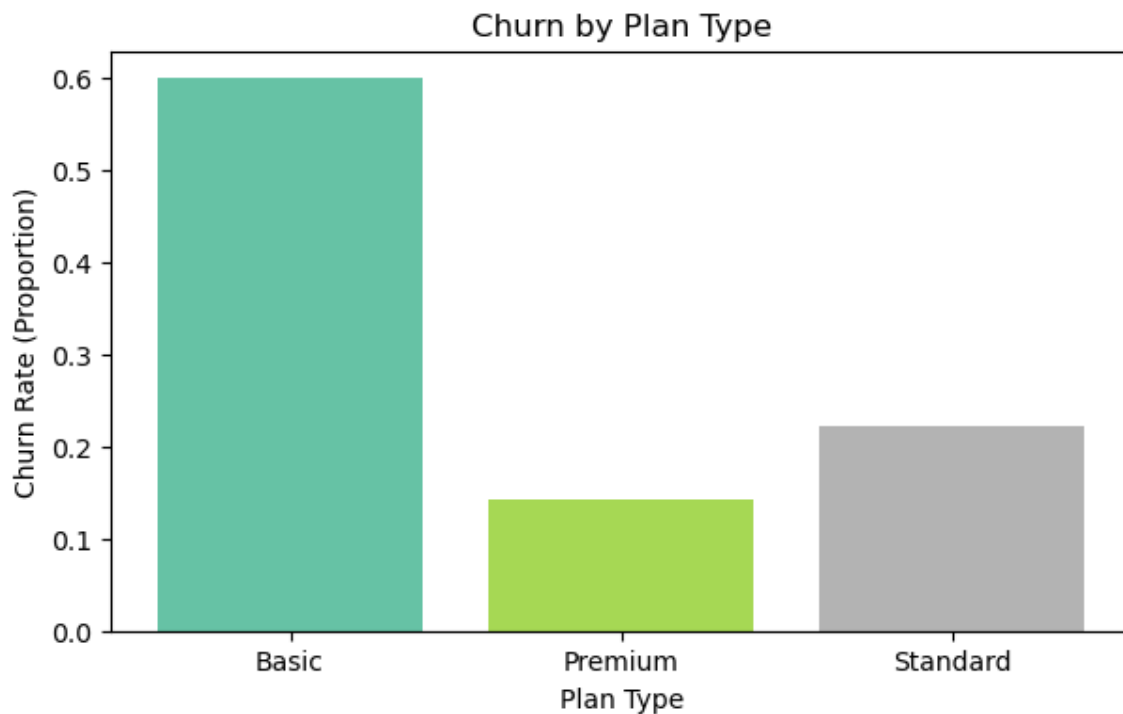


Figure 1. Churn by plan type

The plan-level output shows the highest observed churn rate in Basic at 60.00%, while Premium is 14.29% and Standard is 22.22%.

6. Churn Analysis by State

State	Churn rate
Delhi	25.00%
Karnataka	100.00%
Kathmandu	0.00%
Maharashtra	0.00%
Meghalaya	66.67%
Nagaland	0.00%
Rajasthan	0.00%
Telangana	50.00%
Uttar Pradesh	0.00%

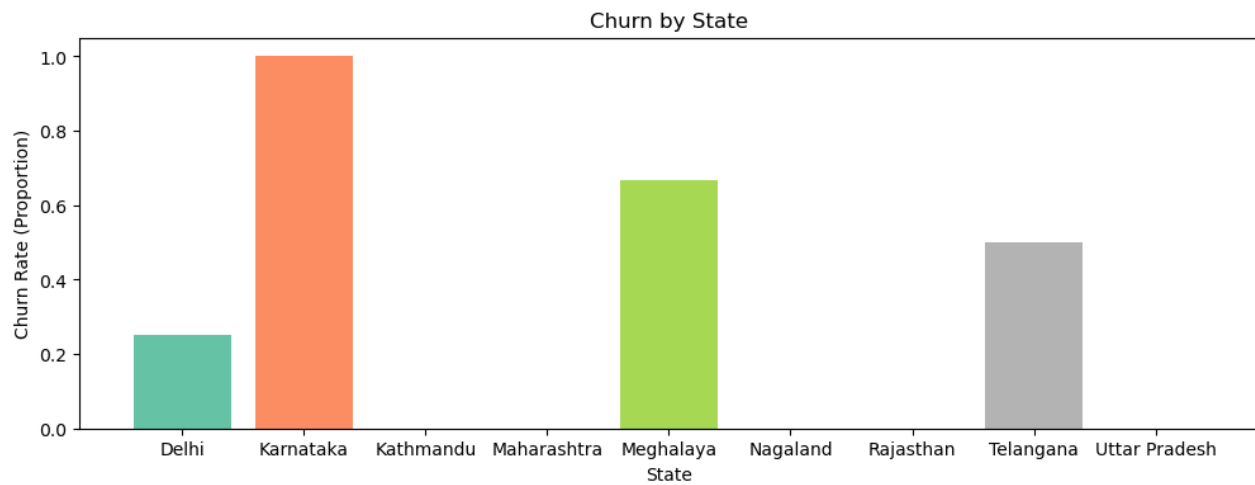


Figure 2. Churn by state

Karnataka records a 100% observed churn rate, Meghalaya has 66.67%, Telangana 50.00%, and Delhi 25.00%. The remaining states show 0% churn.

7. Correlation and Relationship Analysis

The notebook encodes categorical variables and examines relationships using correlation matrix. Priority-based encoding approach is followed.

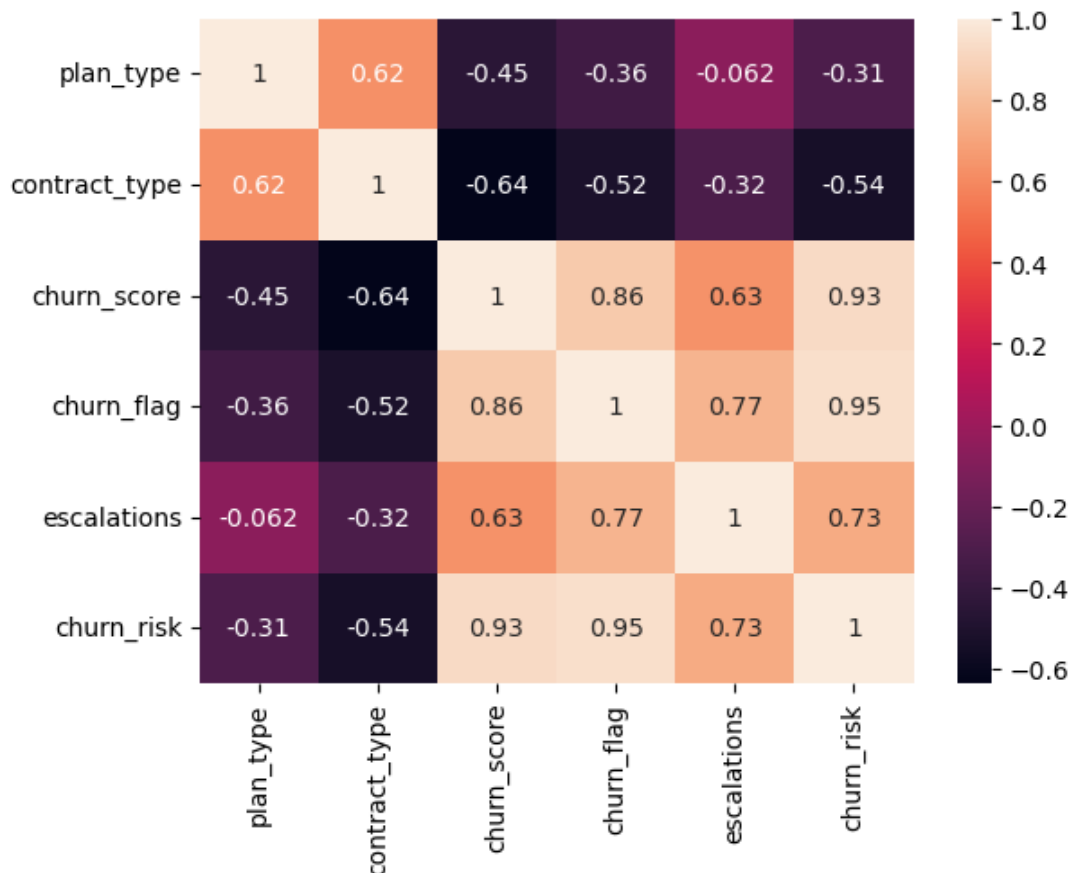


Figure 3. Correlation heatmap with priority-based encoding for ordinal categories.

In the heatmap, churn_flag has a correlation of 0.86 with churn_score, 0.77 with escalations, and 0.95 with churn_risk. churn_score and churn_risk also show a correlation of 0.93.

8. Pairplot: Multivariable Relationships

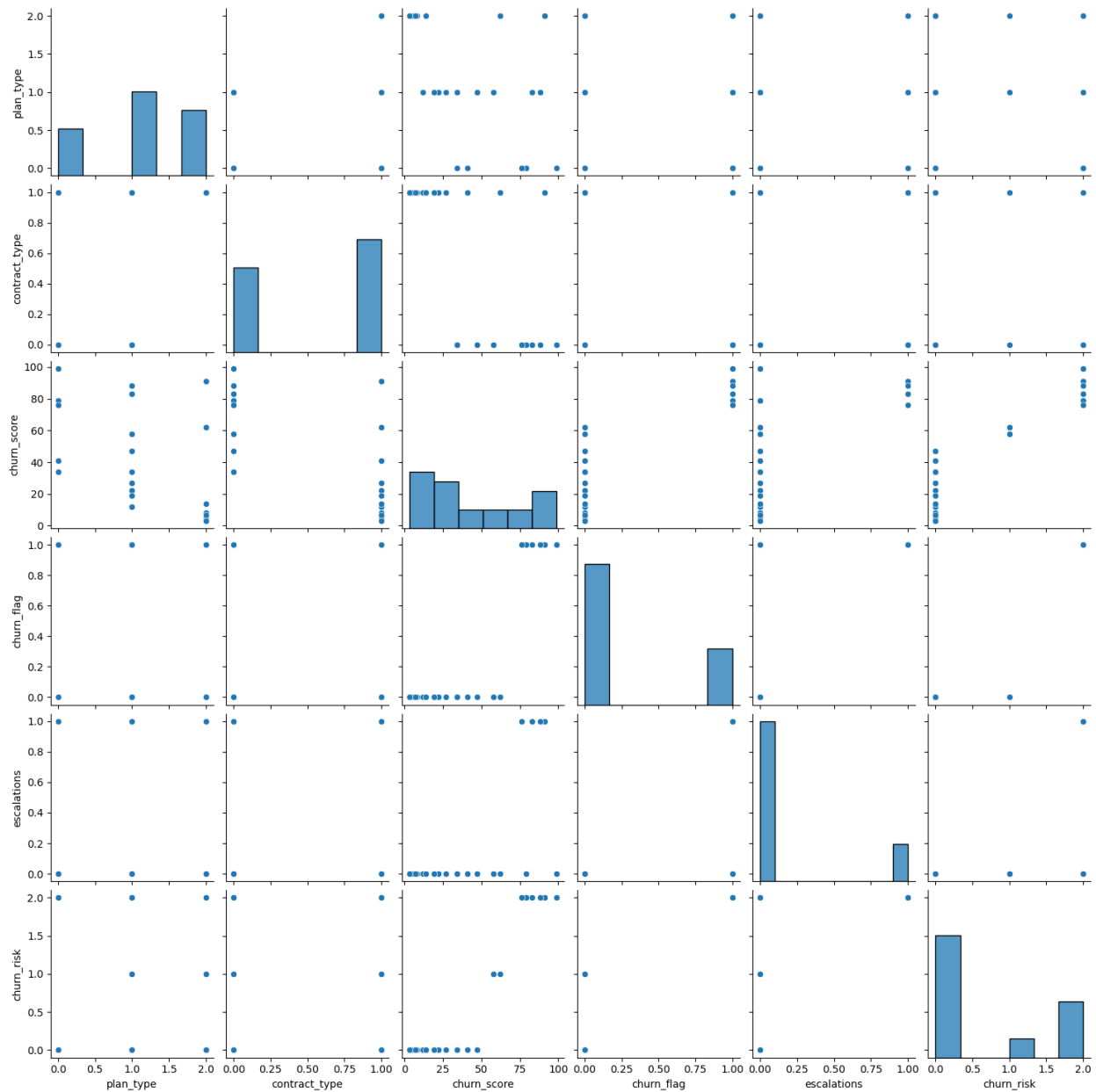


Figure 4. Pairplot of plan type, contract type, churn score, churn flag, escalations and churn risk.

The pairplot provides a compact view of pairwise relationships among the encoded variables.

9. Multi-Dimensional Plan, Charges, Gender and Churn Risk

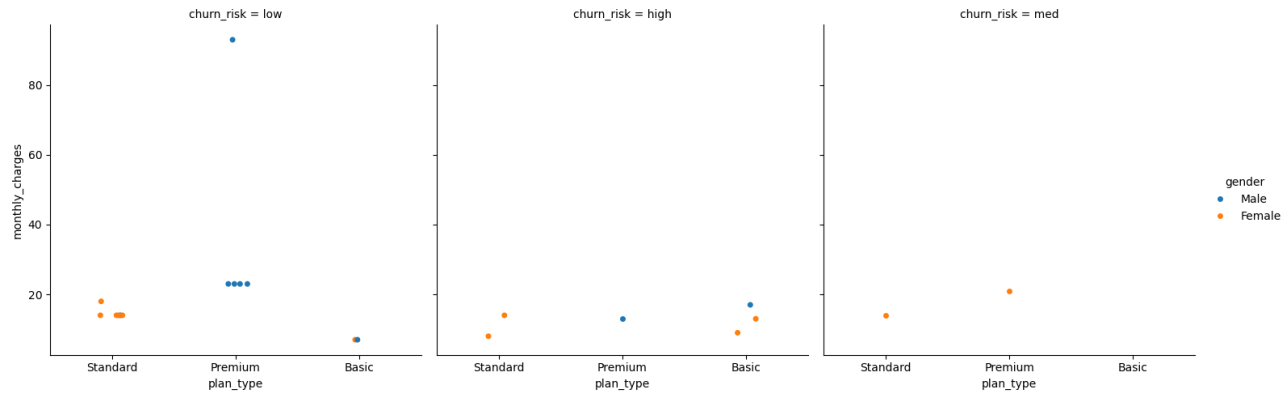


Figure 7. Faceted categorical plot of monthly charges by plan type, split by gender and churn-risk category.

This visualization compares monthly charges across plan types while separating observations into low, medium and high churn-risk facets and using gender as the hue. It is intended as a multi-dimensional exploratory view rather than a standalone causal analysis.

10. Pivot-Table Summary

Pivot-table summary was calculated at plan level.

Plan type	Churn flag mean
Basic	0.600000
Premium	0.142857
Standard	0.222222

11. Findings

- Overall churn in the analyzed customer sample is 28.57%, with 71.43% retained.
- Plan-level churn varies substantially in the sample: Basic 60.00%, Standard 22.22%, Premium 14.29%.
- State-level churn is uneven, with 100.00% in Karnataka, 66.67% in Meghalaya and 50.00% in Telangana.
- Churned customers account for 73.94 (Rs 'K') in monthly charges.
- Escalations occur in 19.05% of the customer records and have a 0.77 correlation with churn_flag in the encoded analysis.
- The churn-risk variable is derived directly from churn_score, so its strong relationship with churn score and churn flag is expected from the construction of the feature.

14. Conclusion

The analysis establishes a complete customer-level churn workflow: data is cleaned across three source tables, support duplicates are consolidated, subscription cancellations are converted into a churn flag, and additional analytical features are created. The resulting analysis shows meaningful variation in churn across plan types, states, etc. while the correlation analysis highlights strong associations among churn, churn score, escalations and the derived churn-risk category.